

Babak Bagheri Hariri

Tehran, Iran

+98 912 238 9146

babakbh@gmail.com

linkedin.com/in/babakbh

github.com/babakbh/ai-agentic-system-samples

Senior AI Engineer | Agentic AI Systems | LLM Applications | Knowledge Graphs

Professional Summary

Senior AI and Software Engineer with 20+ years of experience designing and delivering enterprise software, data-intensive platforms, knowledge graph solutions, and LLM-powered applications.

Experienced in translating complex business requirements into production systems, designing scalable architectures, building AI-assisted applications, and collaborating directly with stakeholders.

Strong background in Python development, knowledge graphs, semantic technologies, Retrieval-Augmented Generation (RAG), and enterprise AI solutions

Core Competencies

- Agentic AI Systems, Large Language Models (LLMs), Generative AI, Retrieval-Augmented Generation (RAG)
- AI-assisted Knowledge Extraction, Information Retrieval, Prompt Engineering, Structured Outputs
- Knowledge Graph Design and Construction, Ontology Engineering, Semantic Modeling (RDF, RDFS, OWL)
- Knowledge-Centric AI Systems combining LLMs, Knowledge Graphs, and Enterprise Data Sources
- Property Graph Technologies, Apache TinkerPop, Gremlin Traversals, RDF-to-LPG Transformation
- Enterprise Software Architecture, Distributed Systems, Backend Engineering, REST APIs
- Data Platforms, Data Integration, Data Modeling, Analytics, Decision Support Systems
- Stakeholder Collaboration, Solution Design, Technical Leadership, Customer-Facing Engineering

Technical Skills

Artificial Intelligence & Agentic Systems:

LLMs, Generative AI, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Prompt Engineering, Structured Outputs, Knowledge-Grounded AI Systems

Knowledge Graphs & Semantic Technologies:

Knowledge Graphs, RDF, RDFS, OWL, SPARQL, Knowledge Representation, Ontology Engineering, Knowledge Extraction, Information Retrieval

Graph Technologies:

Property Graph Modeling (LPG), Apache TinkerPop, Gremlin, RDF-to-Property Graph Transformation

Programming & Software Engineering:

Python, REST APIs, FastAPI, Async Programming, Java, C#, .NET, API Integration, System Design, Software Architecture

Data Engineering & Analytics:

PostgreSQL, MySQL, Redis, Data Pipelines, Data Integration, Data Modeling, Pandas, NumPy

AI-Assisted Development:

ChatGPT, Claude

Professional Experience

Ontopic (Italy)

2022 – 2026

Senior Knowledge Graph & AI Engineer

- Designed and implemented agentic AI workflows using LangGraph for multi-step reasoning, tool orchestration, and ontology-driven schema mapping.
- Built production LLM systems combining Retrieval-Augmented Generation (RAG), knowledge graphs, and semantic reasoning for enterprise knowledge management and information retrieval.
- Developed Python backend services and FastAPI-based APIs supporting AI-powered applications and retrieval pipelines.
- Led the construction of large-scale knowledge graphs from Huawei and Cisco technical documentation, transforming complex network manuals into structured and queryable knowledge assets.
- Implemented multi-agent workflows for schema alignment and ambiguity resolution in ontology mapping tasks.
- Developed evaluation and human-in-the-loop validation workflows to improve accuracy, consistency, and reliability of LLM-based systems in production.
- Designed knowledge-grounded prompting strategies leveraging ontology reasoning and semantic context to improve LLM output quality.
- Integrated AI workflows into Ontopic Studio, enabling human review and refinement of generated mapping proposals.
- Worked with configurable LLM backends (OpenAI, Claude) for flexible experimentation and deployment across environments.
- Collaborated with customers and stakeholders to translate requirements into scalable AI and knowledge-based solutions.

Central Securities Depository of Iran (CSDI)

2018 – 2022

Data Engineer / Project Lead

- Led Project IX, a large-scale data integration initiative for capital market infrastructure.
- Designed and implemented enterprise data pipelines for financial market services.
- Developed scalable data aggregation and processing systems.
- Collaborated with business and technical stakeholders on data-driven financial services.
- Supported operational and analytical data systems for capital market operations.

Digikala

2017 – 2018

Pricing Experience Manager

- Worked at the intersection of pricing systems, marketplace operations, and data analytics.
- Contributed to pricing intelligence and decision-support systems.
- Built analytical approaches for pricing optimization in a large-scale e-commerce platform.
- Collaborated with product and engineering teams on marketplace performance improvements.

Hamkaran System

2015 – 2017

Software Developer

- Developed ERP modules and enterprise backend systems.
- Worked with .NET-based enterprise architectures.
- Applied modular design and dependency injection principles.
- Contributed to enterprise software development lifecycle.

WebNative Software / Toorin Soft

2002 – 2008

Software Developer / Project Manager

- Developed enterprise web applications and backend systems.
- Worked with Java, .NET, and Oracle-based platforms.
- Managed software development projects and client delivery.

Academic Experience

University of Pennsylvania (USA)

2014 – 2016

Postdoctoral Researcher

- Research on Data Provenance and data lifecycle systems.

Education

PhD in Computer Science

2009 – 2013

Free University of Bozen – Bolzano (Italy)

- Focus: Knowledge Representation, Semantic Technologies, Ontologies, Data-Centric Systems, Formal Verification.

M.Sc. in Software Engineering

2006 – 2008

Sharif University of Technology

- Focus: Schema Matching and Ontology Alignment

B.Sc. in Software Engineering

2002 – 2006

Shahid Beheshti University

Research & Publications

- 700+ citations
- h-index: 12
- 20+ publications in AI, data systems, and semantic technologies
- Publications in ACM SIGMOD / PODS, JAIR, ECAI, BPM

EU Research Project

Optique – Scalable End-User Access to Big Data

2012 – 2016

EU FP7 Integrated Project

- Large-scale European research project on data integration and query processing.
- Collaboration with IBM Watson Research and European academic partners.
- Contributions to semantic data access and scalable query systems.